Prior Designs Collection Irrigation Systems

Prior Design #1 - Gabrielle DeStefano

Rain Harvest Systems - Above Ground Rainwater Collection System

Source:

https://www.rainharvest.com/2100-gallon-preconfigured-above-ground-rainwater-collection-system.asp?srsltid=AfmBOoq60-CtbYru4xXfvxxEbSLL42XiNIdOJNR1dMXwU8dHRTJdZjh1

Image:

- images by Rain Harvest Systems

Design:

The "RainFlo 2100AGS" is a system that connects to a downspout on the house by collecting the rainwater that comes off the roof into a 2,100 gallon tank above ground. This is a simple assembly and is designed to reduce time and money spent compared to a ordinary irrigation system. It is a vertical water tank with an external pump that can pump the collected water into hoes or pipes to be able to water a lawn or garden depending on the intended use of the product. "The pump is 1 ½ horsepower and delivers up to 36 GMP" (Rain Harvest Systems). Inside the tank there is a filtration system to filter debris or substances that might have contaminated the rainwater. There is also an overfill system in case of an abundance of rainwater in a short timespan. At the top of the tank there is a rain gauge to measure the amount of water there is in the tank at the time. The tank comes with hardware for assembly, leaf filers, hoes, gauges, a pump, and covers to keep the system out of the elements and appealing to the eye (Rain Harvest Systems).

Analysis:

On this design of a collection method irrigation system there is a pump which might not be ideal for what the customer wants and out of the range of the budget we are allotted. The tank is also very large for what the customer is looking for to supply a 12x12' garden. Although there are some flaws to this design compared to the customer's needs, it is a great design and system that can help support our thoughts and ideas as a guideline of making and improving our collection irrigation system. To make this design suit the needs of the customer we would need to shrink the size of the tank, use gravity instead of a pump, if the budget is low or the customer does not approve, but other than that it is a great design to harvest rainwater for the intended use of watering the home garden. The rain gauge is something that we should look into incorporating into our design to let the customer know how much water is in the tank at the time as well as a filter so that the customer's collected rainwater is clean of debris or substances that might affect the gardener clog the irrigation system.

Prior Design #2 - Ethan Roadfuss

Complete Drip Kit for Rain Barrel Irrigation

Home Depot:
https://www.homedepot.com/p/DIG-Complete-Drip-Kit-for-Rain-Barrel-Irrigation-for-50-Plants-GF100/305188426#overlay
Image:


DIG's GF100 Gravity Feed Drip Irrigation Kit is a complete kit designed for a gravity feed or garden hose system. The kit includes all the parts needed to install a drip irrigation system to water roses, vines, shrubs or vegetables, starting from a raised rain barrel with a hose thread outlet or from a garden faucet using the included pressure regulator. The kit includes a 3/4 in. 155 mesh filter, 3/4 in. swivel adapter, 100 ft. of our premium 1/2 in. drip tubing with 0.700 OD, 50 feet of in. micro tubing, 50 button drippers and all the necessary parts to complete the installation. (Homedepot, 305188426)

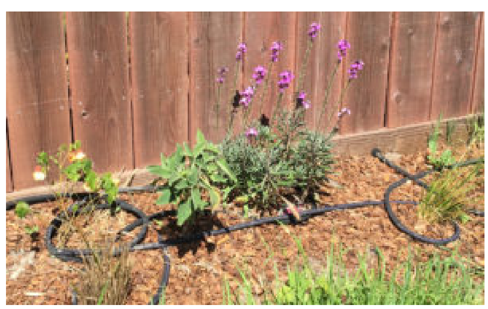
Highlights:
Complete kit including all the parts for a complete system
Button drip emitters turbulent flow through a unique labyrinth water passage reduces likelihood of clogging
Raised barbed outlet prevents water runoff along the drip lateral
Contains all the parts needed to start a system from a 60 to 200 Gal. rain barrel (not included)

Operating pressure controlled by the height of the rain barrel (the higher the barrel height, the higher the flow rate from the drippers)
Constructed of UV-resistant, durable plastic material to withstand the most adverse conditions
Operating pressure: 2 psi to 15 psi
Drip emitter flow rates: 0.5 GPH at 5 psi, 0.8 GPH at 10 psi and 1 GPH at 15 psi (nominal)
Drip emitter inlet size: 1/4 in. barb; drip emitter outlet side: nipple
Poly tubing length and size: 100 ft. x 1/2 ft. with 0.600 ID x 0.700 OD
Micro tubing length size: 50 ft. x 1/4 in.
Total flow rate for this kit: 32 GPH (.53 GPM) (Homedepot, 305188426)

Analysis:
This irrigation system sold by Home Depot is a cost effective and portable system that can hook up to a rain barrel. It can water up to 50 plats and has a filter , normal tubing, and micro tubing. As well as 50 button drippers, so it can sustain enough plants. The 2 - 15 psi means it can be used for smaller and medium sized tanks without being hooked up to power, because it uses gravity. It can use a 60-200 gallon rain barrel.Paired with a low pressure timer, this system could efficiently irrigate a small garden.

Citation: Homedepot. (n.d.). Dig complete drip kit for rain barrel irrigation (for 50 plants) GF100. The Home Depot.
https://www.homedepot.com/p/DIG-Complete-Drip-Kit-for-Rain-Barrel-Irrigation-for-50-Plants-GF100/305188426#overlay

Prior Design #3 - Tez Harris
Blue Barrel Systems - Gravity Feed Irrigation Rain Barrels
Source:
https://www.bluebarrelsystems.com/blog/gravity-feed-irrigate-rain-barrels/
Image:


- images from Jesse (Blue Barrel)

Details:

He uses 4 water cheap and simple giant water tanks, including a small hole at top where he's able to insert water. He uses a water hose that is connected to his tank and has a system connected to it that does a time based control system on when it waters his plants. The hose has many holes going through it where the water exits and water each plant. The hose system he uses has an additional water filter working with it to filter the water.

Analysis:

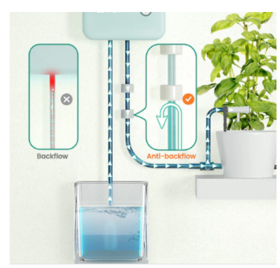
The main unsuccessful thing with this is the ability that each tank has to collect rain with it first being beside the house and under the roof. Second, with each tank having such a small hole how would water really enter? Second to keep animals out of water he could place an metal gated type of material on top only allowing water in.

Prior Design #4 - Gabrielle Stewart

LetPot Automatic Watering System 2.0, Drip Irrigation Kits

Source: LetPot's garden manufactures tools to help with watering plants in a garden. *LetPot Smart Drip Irrigation System*. *LetPot*, letpot.com/products/smart-drip-irrigation-kit?srsltid=AfmBOoqeaMThyh8TLM0mrf6WbJZCvNBSKFHeV7coyqK4VudItH0eEegQ. Accessed 7 Feb. 2025.

Image:

Details:

This system is above-ground. It is also automated so that you can customize when you want your plants to be watered. According to LetPot, "you can tailor the watering intervals within each session by setting the pump's operational and pause durations." The drippers adjust, which allows for different amounts of water to flow to your plant's roots. This system has a filter to prevent any dirt from reaching the plants or damaging the pipes. This system can water up to 20 plants. *LetPot Smart Drip Irrigation System*. *LetPot*, letpot.com/products/smart-drip-irrigation-kit?srsltid=AfmBOoqeaMThyh8TLM0mrf6WbJZCvNBSKFHeV7coyqK4VudItH0eEegQ. Accessed 7 Feb. 2025.

Analysis:

This design solves the problem of irrigation, however, it is hard to collect rainwater and store it with this design.